

Daily Logic Drill #7 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Answers and Methods.

Section A Rapid Recognition

1. To prove statement P by contradiction, you should assume ___ and derive a contradiction.

Answer: The negation of P (i.e. “not P ”).

Method: Proof by contradiction assumes the statement you want to prove is false, then shows that assumption forces something impossible, so the assumption — and hence the negation — must itself be false, leaving P true.

Investigate further: This relies directly on Day 3’s fact (A7) that a true statement’s negation is always false, and vice versa — proof by contradiction is that fact, used as a proof engine.

2. True or false: “Proof by contradiction and proof by contrapositive are the same technique.”

Answer: False.

Method: Contrapositive proves “if P then Q ” by proving “if not Q then not P ” directly — no contradiction is ever reached, it’s a direct proof of an equivalent statement. Contradiction instead assumes not- P (or, for a conditional, P and not- Q together) and derives an impossibility.

Investigate further: Every contrapositive proof *can* be rewritten as a contradiction proof (see A8), but they are not the same technique — contrapositive never needs to name an explicit “impossible” outcome.

3. What contradiction is reached in the classical proof that $\sqrt{2}$ is irrational?

Answer: That p and q are both even, contradicting the assumption that p/q was in lowest terms.

Method: Assuming $\sqrt{2} = p/q$ in lowest terms leads, via $2q^2 = p^2$, to both p and q being even — directly contradicting “lowest terms,” which requires $\gcd(p, q) = 1$.

Investigate further: This is exactly Day 5’s D2 derivation — that question isolated the algebra as correct and only the final English sentence as wrong; this question asks for the contradiction itself.

4. True or false: “If assuming ‘not P ’ leads to a statement that’s always false, P must be true.”

Answer: True.

Method: This is the defining logical principle behind proof by contradiction: since “not P ” cannot coexist with a genuine impossibility, “not P ” must be false, and by A7-style reasoning (Day 3), P is therefore true.

Investigate further: This is the formal statement of what A1 describes procedurally — A1 is “how to do it,” A4 is “why it works.”

5. Give the first line you would write to prove “there are infinitely many primes” by contradiction.

Answer: “Suppose, for contradiction, that there are only finitely many primes.”

Method: Proof by contradiction always opens by assuming the negation of the target statement — here, negating “infinitely many” gives “finitely many.”

Investigate further: The full classical proof (multiply all assumed primes together, add 1, find a new prime factor not on the list) is a famous instance of this exact opening move — naming the right negation is the hardest part.

6. True or false: “Deriving ‘ $1 = 0$ ’ partway through an argument is a valid contradiction to conclude the assumption was false.”

Answer: True.

Method: Any statement that is unconditionally false (like $1 = 0$) is a valid contradiction — it doesn’t need to relate back to the original claim’s subject matter at all, only to be something that can never be true.

Investigate further: This is why classic “proofs” that $1 = 2$ (Day 5, B3) are actually backwards: reaching $1 = 0/1 = 2$ genuinely does signal an error somewhere in the argument (usually an illegal division), exactly as this question implies.

7. True or false: “In a proof by contradiction of ‘if P then Q ’, you assume P is true and Q is false simultaneously.”

Answer: True.

Method: “If P then Q ” is false exactly when P is true and Q is false (its only failure case); assuming that combined case and deriving an impossibility rules it out, leaving the implication true.

Investigate further: This connects directly to Day 3’s fact that “ P and not Q ” is the negation of “if P then Q ” — proof by contradiction of a conditional always starts by assuming exactly that negation.

8. True or false: “Every valid proof by contrapositive can be rewritten as a proof by contradiction.”

Answer: True.

Method: A contrapositive proof shows “not $Q \Rightarrow$ not P ” directly; the same content can always be repackaged as: assume P and not Q (A7’s contradiction setup), use the same reasoning to derive not P , which directly contradicts the assumed P .

Investigate further: This is the precise sense in which the two techniques, while different (A2), are not unrelated — contrapositive is a special, more efficient case of what contradiction can always do.

9. To disprove a universal (“for all”) claim as fast as possible, should you generally reach for a counterexample or for proof by contradiction?

Answer: A counterexample.

Method: Day 3’s quantifier-negation rule says the negation of “for all x , $P(x)$ ” is “there exists x , not

$P(x)$ ” — a single explicit witness. Exhibiting one instance is always faster than setting up and deriving a contradiction.

Investigate further: Contradiction earns its keep proving a claim is *true* (especially existence claims, like “infinitely many primes,” where no single witness could ever suffice) — for disproving, Day 4’s counterexample toolkit is almost always the quicker tool.

10. True or false: “If you can find a single counterexample to a claim, you never need a separate proof by contradiction to disprove that same claim.”

Answer: True.

Method: One genuine counterexample is already a complete disproof of a universal claim (Day 4) — contradiction would only be reaching the same conclusion by a slower, more roundabout route.

Investigate further: This is the practical payoff of A9: always check first whether a quick counterexample exists before reaching for the heavier machinery of contradiction.

Section B Manipulation Drills

1. Prove by contradiction: “There is no smallest positive rational number.”

Answer: Proved.

Method: Suppose, for contradiction, that $q > 0$ is the smallest positive rational. Then $q/2$ is also a positive rational (rationals are closed under division by 2), and $q/2 < q$, contradicting that q was the smallest.

Investigate further: The same halving trick shows there’s no smallest positive real number either — the argument never used that q was specifically rational, only that it was a positive number in a set closed under halving.

2. Prove by contradiction: “If n^2 is even, then n is even.”

Answer: Proved.

Method: Suppose, for contradiction, that n^2 is even but n is odd. Then $n = 2k+1$, so $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd — contradicting that n^2 is even. So n must be even.

Investigate further: This is the exact same fact proved by contrapositive in Day 3 (C2) — comparing the two write-ups shows A8’s claim in action: the contrapositive proof (“if n odd then n^2 odd”) and this contradiction proof share identical algebra, differing only in framing.

3. Prove by contradiction: “There is no largest even integer.”

Answer: Proved.

Method: Suppose, for contradiction, that N is the largest even integer. Then $N + 2$ is also even, and $N + 2 > N$, contradicting that N was the largest.

Investigate further: The structure is identical to B1 (assume an extreme value exists, construct something that beats it) — this “assume an extremum, then beat it” pattern is the most common contradiction template for “there is no largest/smallest” claims.

4. Prove by contradiction: “If a, b are integers and $a + b$ is odd, then a and b cannot both be even.”

Answer: Proved.

Method: Suppose, for contradiction, that $a + b$ is odd and both a, b are even. Write $a = 2j$, $b = 2k$; then $a + b = 2(j + k)$, which is even — contradicting that $a + b$ is odd.

Investigate further: This is Day 1's B8 (“both even $\Rightarrow$ sum even”) used in its contrapositive-flavoured form, but framed here explicitly as a contradiction rather than named as one.

5. Prove by contradiction: “If p is prime and $p > 3$, then p is not divisible by 6.”

Answer: Proved.

Method: Suppose, for contradiction, that $p > 3$ is prime and $6 \mid p$. Then $p = 6k$ for some integer $k \geq 1$, so $2 \mid p$, meaning p has a divisor (2) other than 1 and itself (since $p > 3 > 2$ means $p \neq 2$) — contradicting that p is prime.

Investigate further: This is really the same underlying fact as Day 1's A9/Day 4's A5 (a multiple of 6 is automatically a multiple of 2), now used as the engine of a contradiction rather than stated as a bare implication.

6. Prove by contradiction: “There is no integer n satisfying both $n > 5$ and $n < 3$.”

Answer: Proved.

Method: Suppose, for contradiction, that such an n exists. Then $n > 5$ and $n < 3$ together give $5 < n < 3$, i.e. $5 < 3$ — a numerically false statement, a valid contradiction (A6).

Investigate further: This is the simplest possible contradiction template: the “impossibility” reached is a bare false numerical inequality, needing no algebraic manipulation at all.

7. Prove by contradiction: “If x is a positive real number, then $x + \frac{1}{x} \geq 2$.”

Answer: Proved.

Method: Suppose, for contradiction, that $x > 0$ and $x + \frac{1}{x} < 2$. Multiplying both sides by $x > 0$ (safe, since x is positive — Day 5, C4's exact justification) gives $x^2 + 1 < 2x$, i.e. $x^2 - 2x + 1 < 0$, i.e. $(x - 1)^2 < 0$. But a real square can never be negative — contradiction.

Investigate further: This reuses Day 5's C8 identity $(x - 1)^2 = x^2 - 2x + 1 \geq 0$ directly, just applied to the reversed (assumed-false) inequality to expose the impossibility.

8. Prove by contradiction: “If n is a positive integer, n and $n + 1$ share no common factor greater than 1.”

Answer: Proved.

Method: Suppose, for contradiction, that $d > 1$ divides both n and $n + 1$. Then d divides their difference, $(n + 1) - n = 1$. But no integer $d > 1$ divides 1 — contradiction.

Investigate further: “ d divides a difference of two multiples of d ” is a workhorse fact reused constantly in Number Theory's gcd toolkit (Euclidean algorithm), seen here in miniature.

9. Prove by contradiction: “ $\log_2 3$ is irrational.”

Answer: Proved.

Method: Suppose, for contradiction, that $\log_2 3 = p/q$ for positive integers p, q . Then $2^{p/q} = 3$, so $2^p = 3^q$. But 2^p is even (for $p \geq 1$) while 3^q is odd (for any $q \geq 0$) — an even number cannot equal an odd number, a contradiction.

Investigate further: This is structurally the same “parity mismatch” contradiction as $\sqrt{2}$ ’s irrationality (A3), just swapping the modular obstruction (even vs. odd) for the earlier one (even count of prime factors of 2 vs. odd).

10. Prove by contradiction: “If a, b are positive integers with $a < b$, then $a^2 < b^2$.”

Answer: Proved.

Method: Suppose, for contradiction, that $a < b$ but $a^2 \geq b^2$. Since $a, b > 0$, $a^2 \geq b^2 \iff a \geq b$ (both sides positive, so squaring/rooting preserves the inequality direction) — but this directly contradicts the assumption $a < b$.

Investigate further: This relies on squaring being a strictly increasing, hence order-preserving, function specifically on the positive reals — notably *not* true on all reals (Day 5, B5’s “ $x^2 > 4 \Rightarrow x > 2$ ” flaw is exactly what breaks once negative values are allowed).

Section C Substitution & Structure

1. Prove by contradiction: “ $\sqrt{2} + \sqrt{3}$ is irrational.”

Answer: Proved.

Method: Suppose, for contradiction, that $\sqrt{2} + \sqrt{3} = r$ for some rational r . Then $\sqrt{3} = r - \sqrt{2}$, so squaring both sides gives $3 = r^2 - 2r\sqrt{2} + 2$, i.e. $\sqrt{2} = \frac{r^2 - 1}{2r}$ (valid since $r \neq 0$, as $\sqrt{2} + \sqrt{3} > 0$). The right-hand side is rational (a ratio of rationals), but $\sqrt{2}$ is irrational (Day 5, D2) — contradiction.

Investigate further: The trick of isolating one surd, then squaring to expose the other, is a standard move whenever a sum of two different surds is claimed irrational — naively squaring the whole sum at once leaves a stubborn cross-term ($2\sqrt{6}$) that this rearrangement avoids.

2. Prove by contradiction: “There is no integer solution to $x^2 + y^2 = 4z + 3$.”

Answer: Proved.

Method: Suppose, for contradiction, integers x, y, z satisfy $x^2 + y^2 = 4z + 3$. Every integer square is $\equiv 0$ or $1 \pmod{4}$ (since $(2k)^2 = 4k^2 \equiv 0$ and $(2k+1)^2 = 4k^2 + 4k + 1 \equiv 1$), so $x^2 + y^2 \pmod{4} \in \{0, 1, 2\}$. But $4z + 3 \equiv 3 \pmod{4}$, which is not in $\{0, 1, 2\}$ — contradiction.

Investigate further: This “sum of two squares mod 4” obstruction is a classical tool for ruling out entire families of Diophantine equations at once, without checking any individual case.

3. You are given the identity $2^n - 1 = (2^a - 1)(2^{n-a} + 2^{n-2a} + \dots + 1)$ whenever a divides n . Using proof by contradiction, show that if $2^n - 1$ is prime then n is prime.

Answer: Proved.

Method: Suppose, for contradiction, that $2^n - 1$ is prime but n is not prime. Since $n > 1$ (as $2^1 - 1 = 1$ isn’t prime), n composite means $n = ab$ for integers $1 < a, b < n$. Taking a as a proper divisor of n with $1 < a < n$, the given identity factors $2^n - 1$ as $(2^a - 1)$ times a second factor, with $2^a - 1 \geq 2^2 - 1 = 3 > 1$ and the second factor also > 1 (a sum of at least two positive terms). This exhibits $2^n - 1$ as a product of two integers each greater than 1 — contradicting that $2^n - 1$ is prime.

Investigate further: This is T from Day 4/Day 6's D5, finally proved rather than just cited — Day 4 established the fact was true and explored its converse; this is the actual argument underneath it.

4. Prove by contradiction: “If n is a positive integer that is not a perfect square, then $\sqrt{n}$ is irrational.”

Answer: Proved.

Method: Suppose, for contradiction, that n is not a perfect square but $\sqrt{n} = p/q$ in lowest terms for positive integers p, q . Then $nq^2 = p^2$. Every prime factor of q appears to an even power in q^2 , hence in nq^2 ; since $nq^2 = p^2$, every prime factor of n must also combine with q^2 's contribution to reach an even power overall. Because $\gcd(p, q) = 1$, no prime divides both p and q , so $q = 1$ is forced (any prime dividing q would have to divide p too, to balance $nq^2 = p^2$'s factorisation) — meaning $\sqrt{n} = p$ is an integer, so $n = p^2$ is a perfect square. This contradicts the assumption that n is not a perfect square.

Investigate further: This generalises the $\sqrt{2}$ argument (A3) — $\sqrt{2}$ irrational is just the special case $n = 2$ — to every non-square positive integer at once, using prime factorisation instead of a single parity check.

5. Using proof by contradiction and the fact that 2 is the only even prime (Day 1, B9), prove there do not exist two distinct primes p, q , both greater than 2, with $p + q = 5$.

Answer: Proved.

Method: Suppose, for contradiction, that distinct primes $p, q > 2$ exist with $p + q = 5$. Since $p, q > 2$, both are odd (the only even prime is 2, Day 1's B9). But an odd plus an odd is even, so $p + q$ is even — contradicting $p + q = 5$, which is odd.

Investigate further: This uses B9's fact for at least the fourth distinct purpose across this pillar (Day 1's B9 itself, Day 4's A4/B9, Day 5's C7, and now here) — a genuinely load-bearing fact worth having ready without re-deriving.

6. Using contradiction and the fact that $n^2 - n$ is always even (Day 1, D5 / Day 6, C5), prove there is no positive integer n for which $n^2 - n = 7$.

Answer: Proved.

Method: Suppose, for contradiction, that a positive integer n satisfies $n^2 - n = 7$. Since $n^2 - n = n(n - 1)$ is a product of two consecutive integers, it is always even (Day 1, D5). But 7 is odd — contradicting $n^2 - n = 7$.

Investigate further: No factoring or quadratic formula was needed at all — the parity obstruction alone rules out every integer n simultaneously, far faster than solving $n^2 - n - 7 = 0$ directly.

7. Prove by contradiction: “The equation $x^2 + y^2 = 3$ has no solution in positive integers.”

Answer: Proved.

Method: Suppose, for contradiction, positive integers x, y satisfy $x^2 + y^2 = 3$. Since $x, y \geq 1$, both $x^2, y^2 \geq 1$. If either $x \geq 2$ or $y \geq 2$, that term alone is $\geq 4 > 3$, already too large. So $x = y = 1$ is forced, giving $x^2 + y^2 = 2 \neq 3$ — contradiction.

Investigate further: This is a bounding argument (checking that the only candidates small enough to fit are $x = y = 1$, then testing it) rather than a modular one — both C2 and C7 disprove a Diophantine equation, but by genuinely different mechanisms.

8. Using contradiction, prove that if a, b, c are positive integers with $a^2 + b^2 = c^2$ and c is even, then a and b cannot both be odd.

Answer: Proved.

Method: Suppose, for contradiction, that a, b are both odd and c is even with $a^2 + b^2 = c^2$. An odd square is $\equiv 1 \pmod{4}$ (Day 3, C2), so $a^2 + b^2 \equiv 1 + 1 = 2 \pmod{4}$. But c even means $c = 2k$, so $c^2 = 4k^2 \equiv 0 \pmod{4}$. Since $2 \not\equiv 0 \pmod{4}$, this contradicts $a^2 + b^2 = c^2$.

Investigate further: This reuses Day 3's C2 fact (" $\text{odd}^2 \equiv 1 \pmod{8}$ ", which in particular gives $\equiv 1 \pmod{4}$) to rule out an entire parity pattern in Pythagorean triples at once, without needing any specific triple.

Section D Challenge

1. Consider: I. "Proof by contradiction and proof by contrapositive always produce identical proofs." II. "A single counterexample is a special case of proof by contradiction." III. "If a statement's negation leads to ' $1 = 0$ ', the statement is proven true." Which are true?

Answer: C) III only.

Method: I is false (A2: they are different techniques, even though A8 shows contrapositive can always be *rewritten as* contradiction — "always producing identical proofs" is a stronger, false claim). II is false: a counterexample directly falsifies a universal claim by exhibiting a witness (Day 4); it doesn't assume a negation and derive an impossibility, so it isn't itself an instance of contradiction, even though both are disproof tools. III is true (A6).

Investigate further: II is a tempting near-miss: counterexamples and contradiction both disprove things, but only one of them works by assuming a negation and finding an impossibility — conflating "both are disproof methods" with "one is a case of the other" is the trap.

2. Prove, using proof by contradiction, that there is no positive integer n for which both n and $n + 1$ are perfect squares.

Answer: Proved.

Method: Suppose, for contradiction, that $n = a^2$ and $n + 1 = b^2$ for positive integers a, b . Then $b^2 - a^2 = 1$, i.e. $(b - a)(b + a) = 1$. Since a, b are positive integers with $b > a$ (as $b^2 > a^2$), both $b - a$ and $b + a$ are positive integers, and their product is 1, forcing $b - a = 1$ and $b + a = 1$. But $b - a = 1$ and $b + a = 1$ together give $a = 0$ — contradicting that $n = a^2$ is a positive integer (which requires $a \geq 1$).

Investigate further: This is B8's "consecutive integers share no common factor > 1 " idea in a different guise: two consecutive perfect squares differ by exactly 1, and factoring that difference of squares leaves essentially no room to be positive integers.

3. In a room of at least 2 people, every person is either a knight (always tells the truth) or a knave (always lies). Every person in the room says: "Everyone else in this room is a knave." Using proof by contradiction, determine exactly how many knights are in the room.

Answer: Exactly 1 knight.

Method: Suppose, for contradiction, there are 0 knights (everyone is a knave). Then any one knave's statement "everyone else is a knave" would be entirely true — but knaves never tell the truth, contradiction. Suppose instead there are ≥ 2 knights. Take any one of these knights: their statement "everyone else is a knave" must be true, but a second knight in the room is not a knave — contradiction. So the number

of knights is neither 0 nor ≥ 2 , leaving exactly 1.

Investigate further: This is Day 3's D3 and Day 6's D2 puzzles generalised to an arbitrary room size — the method (rule out each alternative in turn via contradiction) scales even when the room size itself isn't fixed in advance.

4. Using proof by contradiction, prove that the equation $n^2 = 2^k$ has no solution in positive integers n, k with k odd.

Answer: Proved.

Method: Suppose, for contradiction, positive integers n, k satisfy $n^2 = 2^k$ with k odd. Write n 's prime factorisation as $n = 2^m \times (\text{odd part})$; since $n^2 = 2^k$ has no odd prime factors, n 's odd part must be 1, so $n = 2^m$ for some non-negative integer m . Then $n^2 = 2^{2m}$, so $2^{2m} = 2^k$, forcing $k = 2m$ — an even number. This contradicts the assumption that k is odd.

Investigate further: This “exponent of 2 must double when you square” fact is the engine behind the classical proof that $\sqrt{2}$ is irrational when phrased via prime factorisation (C4's method) — D5 makes that connection explicit.

5. Give a second, different proof that $\sqrt{2}$ is irrational (Day 5, D2 gave one proof already), this time using the same 2-adic valuation technique as D4.

Answer: Proved.

Method: Suppose, for contradiction, $\sqrt{2} = p/q$ for positive integers p, q . Then $p^2 = 2q^2$. Let k be the exponent of 2 in p^2 's prime factorisation; by D4's reasoning, since p^2 is a perfect square, k must be even. Now consider $2q^2$: the exponent of 2 here is (exponent of 2 in q^2 , which is even by the same squaring argument) $+1$ (from the explicit factor of 2 out front), which is therefore odd. So $p^2 = 2q^2$ requires an even exponent of 2 on the left to equal an odd exponent of 2 on the right — contradiction, since a single integer has one well-defined prime factorisation.

Investigate further: This closes the pillar exactly where Day 5's D2 opened the topic: the same irrationality fact, now proved by counting prime factors of 2 (an argument that generalises instantly to $\sqrt{p}$ for any prime p , unlike the original parity-only argument) rather than by the original's direct “both p, q even” descent.